

UST'-ISHIM, Russian Federation

WMO: 283820

Lat: 57.683N

Lon: 71.183E

Elev: 56

StdP: 100.66

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -36.5	(c) -33.6	(d) -39.6	(e) 0.1	(f) -36.4	(g) -36.7	(h) 0.1	(i) -33.5	(j) 8.0	(k) -10.0	(l) 7.1	(m) -9.1	(n) 0.9	(o) 280	(p) 0.646

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.4	(c) 28.7	(d) 20.2	(e) 27.0	(f) 19.2	(g) 25.3	(h) 18.2	(i) 21.4	(j) 26.7	(k) 20.4	(l) 25.2	(m) 19.3	(n) 23.7	(o) 3.3	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.6	14.4	24.3	18.6	13.5	22.9	17.5	12.6	21.7	62.5	26.5	58.9	25.2	55.3	23.8	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min		Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.3	7.4	6.5		-39.4	31.6	3.7	1.6	-42.1	32.8	-44.3	33.7	-46.4	34.6	-49.2	35.8
				DB				-42.3	24.2	-44.4	25.1	-46.5	25.9	-49.2	26.9
				WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.8	-19.0	-16.1	-6.9	2.2	10.5	16.5	18.0	15.3	9.3	2.2	-8.1	-15.4		
	DBStd	14.31	8.39	7.76	6.13	5.57	5.16	4.65	3.89	3.81	4.29	4.94	7.96	8.69		
	HDD10.0	4099	899	730	523	238	59	6	1	3	65	243	543	788		
	HDD18.3	6498	1158	963	782	484	247	87	56	110	272	500	793	1047		
	CDD10.0	743	0	0	0	4	75	202	249	167	43	2	0	0		
	CDD18.3	98	0	0	0	0	4	33	46	15	1	0	0	0		
	CDH23.3	855	0	0	0	1	81	292	352	119	9	0	0	0		
	CDH26.7	180	0	0	0	0	18	67	73	21	0	0	0	0		
(14) Wind	WSAvg	2.9	2.4	2.5	3.2	3.4	3.4	2.9	2.7	2.7	2.9	3.1	3.1	2.7		
(15) Precipitation	PrecAvg	490	23	17	22	27	42	58	73	69	51	41	37	30		
	PrecMax	724	40	37	55	55	102	143	220	146	90	104	70	61		
	PrecMin	307	3	1	0	1	8	10	16	23	14	3	7	9		
	PrecStd	87	12	7	13	14	19	29	47	27	22	23	16	12		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.8	2.4	7.5	21.2	28.8	30.8	30.7	29.3	24.7	17.0	6.1	1.8		
		MCWB	0.0	0.4	3.0	11.7	16.2	19.6	21.7	21.0	16.0	10.8	4.4	1.2		
	2%	DB	-1.8	-0.1	4.6	17.0	25.4	28.7	28.5	26.3	21.2	13.6	4.1	0.3		
		MCWB	-2.5	-1.2	1.4	10.1	14.9	19.9	20.8	19.2	14.8	9.2	2.9	-0.2		
	5%	DB	-3.9	-2.1	2.9	13.5	22.9	26.7	27.1	24.3	18.5	11.2	2.4	-1.8		
		MCWB	-4.5	-3.0	0.5	7.8	14.0	19.0	19.9	18.4	13.5	8.0	1.5	-2.5		
	10%	DB	-6.4	-4.5	1.5	10.9	20.0	24.7	25.2	22.2	16.2	9.2	1.0	-3.7		
		MCWB	-7.0	-5.5	-0.4	6.3	12.5	18.0	19.0	17.4	12.3	6.8	0.3	-4.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.2	0.7	3.5	12.4	17.4	22.7	22.9	21.7	17.1	11.5	4.9	1.3		
		MCDB	0.8	2.0	6.4	19.7	25.9	27.5	28.6	27.6	21.8	15.4	5.9	1.6		
	2%	WB	-2.4	-1.2	1.9	10.3	16.0	21.0	21.7	20.2	15.5	9.8	3.2	-0.3		
		MCDB	-1.8	-0.2	3.9	15.9	23.9	26.5	27.0	24.7	20.1	12.6	4.1	0.2		
	5%	WB	-4.5	-3.0	0.9	8.4	14.6	19.8	20.7	19.0	14.0	8.4	1.5	-2.3		
		MCDB	-4.0	-2.2	2.6	13.0	21.3	25.0	25.6	23.0	17.6	10.6	2.2	-1.8		
	10%	WB	-6.9	-5.4	-0.2	6.5	13.2	18.6	19.7	17.8	12.7	7.0	0.4	-4.3		
		MCDB	-6.4	-4.6	1.4	10.4	18.7	23.5	24.0	21.5	15.6	9.3	1.0	-3.7		
(35) Mean Daily Temperature Range	MDBR		8.6	10.8	10.9	10.2	11.5	10.2	9.4	9.0	8.4	6.6	6.2	7.9		
	5% DB	MCDBR	7.7	8.4	10.1	14.6	15.9	13.5	12.1	12.2	12.4	10.6	4.8	6.4		
		MCWBR	7.6	7.6	7.6	8.6	7.2	6.0	5.6	5.9	6.5	6.9	4.4	6.2		
	5% WB	MCDBR	7.8	8.4	8.6	13.6	14.2	11.8	10.9	11.0	11.1	9.8	4.8	6.2		
		MCWBR	7.7	7.7	6.8	8.6	7.3	6.0	5.5	5.9	6.4	6.8	4.5	6.2		
(40) Clear-Sky Solar Irradiance	taub		0.275	0.311	0.369	0.408	0.403	0.409	0.427	0.423	0.375	0.349	0.306	0.275		
	taud		2.021	2.015	1.929	2.050	2.252	2.304	2.251	2.232	2.324	2.243	2.086	1.969		
	Ebn at Noon		541	692	745	791	830	830	802	774	758	656	495	404		
	Edh at Noon		58	93	135	141	124	119	123	117	92	75	55	46		
(44) All-Sky Solar Radiation	RadAvg		0.68	1.67	3.12	4.45	5.18	5.54	5.32	4.03	2.64	1.31	0.59	0.39		
	RadStd		0.06	0.13	0.22	0.37	0.41	0.43	0.53	0.34	0.27	0.17	0.11	0.05		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page